Mediatrix®

EMPOWERING THE EDGE OF THE IP NETWORK

Unit Manager Express

User's Manual

Product Version 1.2

Document Revision A

February 8, 2005

Mediatrix Telecom, Inc.
4229 Garlock Street
Sherbrooke, Québec, Canada J1L 2C8

Unit Manager Express Express User's Manual

© 2001-2005, Mediatrix Telecom, Inc.

All rights reserved. No part of this publication may be reproduced or used in any form or by any means – graphic, electronic, or mechanical, including photocopying, recording, taping, or information storage and retrieval systems – without the express written permission of the publisher.

Mediatrix Telecom, Inc. reserves the right to revise this publication and make changes at any time and without the obligation to notify any person and/or entity of such revisions and/or changes.

Trademarks

Microsoft and Windows are registered trademarks of Microsoft Corporation.

Adobe and Acrobat are registered trademarks of Adobe Systems Incorporated.

All other trademarks and registered trademarks are the property of their respective owners.

Contents

Preface

Welcome v

Intended Audience v
Using this Manual vi
Symbols vi
Screen Captures vii
Moving Through the Unit Manager Express vii
SCN vs PSTN viii
End User Technical Support viii

Chapter 1

Installing the Unit Manager Express 1

Platform Requirements 1
Installing the Unit Manager Express 1
Modifying an Existing Installation 2
Installing a MIB File 3
Uninstalling the Unit Manager Express 4

Chapter 2

Introduction 5

What is SNMP? 5
SNMP Messages 6
Definitions 7
Changing a Parameter Value 8

Chapter 3

Using the Unit Manager Express 9

Unit Manager Express Main Window 9
Toolbar 11
Setting SNMP Preferences 12
GetBulk Settings 13
SNMPv3-Security Settings 14
Connecting to a Remote Unit 16
Autodetecting New Units 17

SNMPv3 Unit Settings 19
Opening a MIB File 23
MIB File Icons 24
Notes on MIB files 24
Performing SNMP Operations 26
Performing a GET Operation 26
Performing a SET Operation 27
Performing a SNMP Walk Operation 29
SNMP Table Viewer 30
Options 30
Performing a GET Operation 31
Performing a SET Operation 31
Performing a Walk Operation 31
Generating a XML Configuration File 32
Preferences 32
Generating the File 33
Miscellaneous Options 36
Using the Find Option 36
Expanding and Collapsing the MIB Tree 36
Message Log 37

Chapter 4

Using Scripts 39

Recording a Script 39
Running a Script 41
Editing a Script 42
Generating a Script 43

Appendix A

Glossary 45

Appendix B

List of Acronyms 47

PREFACE

P

Welcome

Thank you for using the Unit Manager Express from Mediatrix Telecom, Inc. The Unit Manager Express can easily detect and configure remote APA units that use the SNMP protocol. It can also manage any other device that uses the SNMPv1, SNMPv2c, or SNMPv3 protocol, no matter what the signalling protocol used.

The Unit Manager Express allows to perform the SNMP GET and SNMP SET operations on MIB variables. It contains features such as SNMP Table viewer, real-time graphical presentation of queried numerical values, etc.

This manual explains how to install and use the Unit Manager Express in conjunction with Mediatrix Telecom, Inc.'s APA products.

Intended Audience

This manual provides all the information you need to install and use the Unit Manager Express. It is intended for network administrators who are responsible for installing and setting up network equipment; consequently, it assumes a basic working knowledge of LANs (Local Area Networks).

What's New in this Version

- Possibility to generate a SML configuration file from the unit's configuration. See "Generating a XML Configuration File" on page 32 for more details.
- Possibility to generate a script. See "Generating a Script" on page 43 for more details.

Using this Manual

This user's manual includes task-related information to help you use the Unit Manager Express as quickly as possible.

Symbols

The following information provides an explanation of the symbols which appear in the Unit Manager Express documentation.

Warning: Means danger. You are in a situation that could cause bodily injury. Before you work on any equipment, you must be aware of the hazards involved with electrical circuitry and familiar with standard practices for preventing accidents.

Waarschuwing: Dit waarschuwingssymbool betekent gevaar. U overtreat in een situatie die lichamelijk letsel kan veroorzaken. Voordat u aan enige apparatuur gaat werken, dient u zich bewust te zijn van de bij elektrische schakelingen betrokken risico's en dient u op de hoogte te zijn van standaard maatregelen om ongelukken te voorkomen.

Varoitus: Tämä varoitusmerkki merkitsee vaaraa. Olet tilanteessa, joka voi johtaa ruumiinvammaan. Ennen kuin työskentelet minkään laitteiston parissa, ota selvää sähkökytkentöihin liittyvistä vaaroista ja tavanomaisista onnettomuuksien ehkäisykeinoista.

Attention: Ce symbole d'avertissement indique un danger. Vous vous trouvez dans une situation pouvant causer des blessures ou des dommages corporels. Avant de travailler sur un équipement, soyez conscient des dangers posés par les circuits électriques et familiarisez-vous avec les procédures couramment utilisées pour éviter les accidents.

Warnung: Dieses Warnsymbol bedeutet Gefahr. Sie befinden sich in einer Situation, die zu einer Körperverletzung führen könnte. Bevor Sie mit der Arbeit an irgendeinem Gerät beginnen, seien Sie sich der mit elektrischen Stromkreisen verbundenen Gefahren und der Standardpraktiken zur Vermeidung von Unfällen bewußt.

Avvertenza: Questo simbolo di avvertenza indica un pericolo. La situazione potrebbe causare infortuni alle persone. Prima di lavorare su qualsiasi apparecchiatura, occorre conoscere i pericoli relativi ai circuiti elettrici ed essere al corrente delle pratiche standard per la prevenzione di incidenti.

Advarsel: Dette varselsymbolet betyr fare. Du befinner deg i en situasjon som kan føre til personskade. Før du utfører arbeid på utstyr, må du vare oppmerksom på de faremomentene som elektriske kretser innebærer, samt gjøre deg kjent med vanlig praksis når det gjelder å unngå ulykker.

Aviso: Este símbolo de aviso indica perigo. Encontra-se numa situação que lhe poderá causar danos físicos. Antes de começar a trabalhar com qualquer equipamento, familiarize-se com os perigos relacionados com circuitos eléctricos, e com quaisquer práticas comuns que possam prevenir possíveis acidentes.

¡Advertencia!: Este símbolo de aviso significa peligro. Existe riesgo para su integridad física. Antes de manipular cualquier equipo, considerar los riesgos que entraña la corriente eléctrica y familiarizarse con los procedimientos estándar de prevención de accidentes.

Varning!: Denna varningssymbol signalerar fara. Du befinner dig i en situation som kan leda till personskada. Innan du utför arbete på någon utrustning måste du vara medveten om farorna med elkretsar och känna till vanligt förfarande för att förebygga skador.

Caution: Caution indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury and/or damage to the equipment or property.

Note: Note indicates important information about the current topic.

▶ **This represents steps you must perform to accomplish a particular task.**

Screen Captures

This user's manual includes sample screen captures. Your actual screen can look slightly different from the sample screen. This is normal and not a cause for concern.

Moving Through the Unit Manager Express

Moving through the Unit Manager Express is like moving through any other Windows program. You can use your mouse to point and click or you can use the <Tab> key to move from field to field.

You can also use short-cut keys. These keys are used in menus and dialog boxes where an alternate keystroke sequence is available. These shortcut keys are shown underscored like Help. To use this shortcut key, press and hold the <ALT> key and press the <H> key. For more information about using shortcut keys and other Windows conventions, see the Microsoft Windows help.

SCN vs PSTN

In Mediatrix Telecom, Inc.'s and other vendor's documentation, the terms SCN and PSTN are used. A SCN (Switched Circuit Network) is a general term to designate a communication network in which any user may be connected to any other user through the use of message, circuit, or packet switching and control devices. The Public Switched Telephone Network (PSTN) or a Private Branch eXchange (PBX) are examples of SCNs.

End User Technical Support

In order to maximize technical support resources, Mediatrix Telecom, Inc. works through its partner channels to resolve technical support issues. All end users requiring technical support are encouraged to contact their vendor directly.

CHAPTER 1

Installing the Unit Manager Express

This chapter describes how to install the Unit Manager Express and its different modules.

Platform Requirements

The Unit Manager Express can be installed on a PC running one of the following operating systems:

- Windows 95
- Windows 98
- Windows NT Workstation
- Windows NT Server
- Windows 2000
- Windows XP

Installing the Unit Manager Express

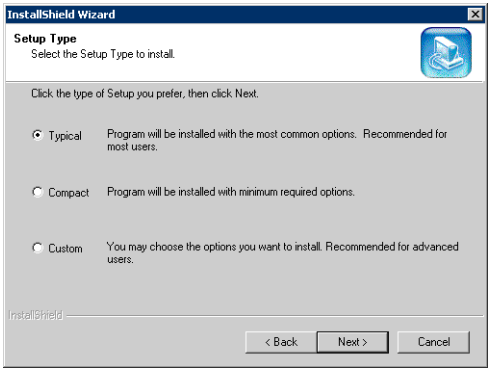
A graphical Installation program will help you quickly install your copy of the Unit Manager Express.

To install the Unit Manager Express:

1. Choose a Windows machine that will host the Unit Manager Express.
2. In the directory where the installation program is located, select the *setup.exe* file and start it. The installation procedure starts.
3. After reviewing the instructions in the Welcome dialog box, keep the preset installation folder, or click *Browse* and then specify the pathname of a different folder.

   The following window opens.

Figure 1: Installation Wizard



4. Select the *Typical* type of installation, then click *Next*.

   The installation starts. Follow the instructions on screen. When the installation is complete, you can run the Unit Manager Express.

Modifying an Existing Installation

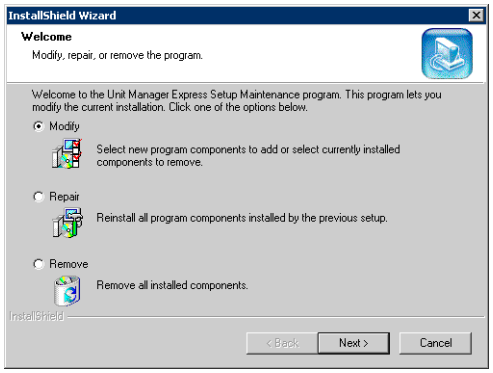
You can modify a previous installation of the Unit Manager Express.

▶ **To modify an already installed version of the Unit Manager Express:**

1. In the directory where the installation program is located, select the *setup.exe* file and start it.

   The installation procedure starts and the following window opens:

Figure 2: Modifying an Installation



You have the following choices:

Table 1: Installation Wizard Choices

Choice	Description
Modify	You can select new program components to add or select currently installed components to remove.
Repair	Reinstalls all program components installed by the previous setup.
Remove	Removes all installed components.

2. Proceed with the rest of the procedure.

Installing a MIB File

The MIB file contains the actual variables you will be able to set for your units. You can install the specific MIB file that you require. This file will be automatically opened when you start the Unit Manager Express.

▶ **To install a MIB file:**

1. Start the MibInstaller by double-clicking the installer file.

   The installation procedure starts.

2. Follow the instructions on screen.

   The MIB file will be installed in the *Mibs* directory under the *Unit Manager Express* installation directory. If you install more than one MIB file, the last one installed will be automatically opened in the Unit Manager Express.

   For more details on MIB files, see "SNMPv3 Unit Settings" on page 19.

Uninstalling the Unit Manager Express

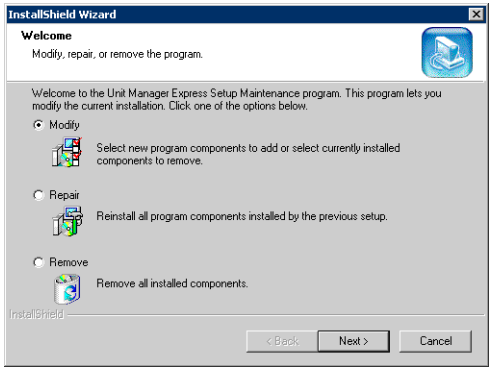
Uninstalling the Unit Manager Express is an easy task to perform.

▶ **To uninstall the Unit Manager Express:**

1. Backup any files if this is only a temporary uninstall.
2. In the directory where the installation program is located, select the *setup.exe* file and start it. The installation procedure starts.

   The following window opens:

   Figure 3: Uninstalling the Unit Manager Express



3. Select the *Remove* option, then click *Next*.

   This will remove all installed components of the Unit Manager Express.
4. Follow the instructions on screen.

CHAPTER 2

Introduction

This chapter introduces the concept of SNMP (Simple Network Management Protocol) and MIB (Management Information Base) files. Using the Unit Manager Express requires a knowledge of MIBs and SNMP. If you do not know SNMP, Mediatrix Telecom, Inc. recommends that you familiarise yourself with it before attempting to modify the variables in the MIB.

What is SNMP?

SNMP is a simple request-reply protocol for Internet network management services. It consists of *network management stations* communicating with *network elements*. Management stations are normally workstations that display relevant facts about the elements being monitored.

SNMP works over the IP (Internet Protocol) communication stack. SNMP network management consists of three pieces:

1. The protocol between the manager and the element, called the *Simple Network Management Protocol* (SNMP). This details the format of the packets exchanged. Although a wide variety of transport protocols could be used, UDP is normally used with SNMP.
2. A set of common structures and an identification scheme used to reference the variables in the MIB. This is called the *Structure of Management Information* (SMI).
3. A *Management Information Base* (MIB) that specifies what variables the network elements maintain (the information that can be queried and set by the manager). See "Management Information Base (MIB)" on page 7 for more details.

SNMP Messages

The Unit Manager Express supports three (3) versions of SNMP: SNMPv1, SNMPv2c, and SNMPv3. SNMP defines a few types of messages that are exchanged between the manager and agent.

SNMPv1

SNMPv1 is the first version of the protocol. The following messages are specific to SNMPv1.

Table 2: SNMPv1 Message Types

Operator	Description
get-request	Get the value of one or more variables.
get-next-request	Get the next variable after one or more specified variables.
set-request	Set the value of one or more variables.
get-response	Return the value of one or more variables. This is the message returned by the agent to the manager in response to the **get-request**, **get-next-request**, and **set-request** operators.
trap	Notify the manager when something happens on the agent.

The first three messages are sent from the manager to the agent, and the last two are from the agent to the manager.

SNMPv2c

There are a few flavours of SNMPv2, SNMPv2c being the most common. The following message is specific to SNMPv2.

Table 3: SNMPv2 Message Type

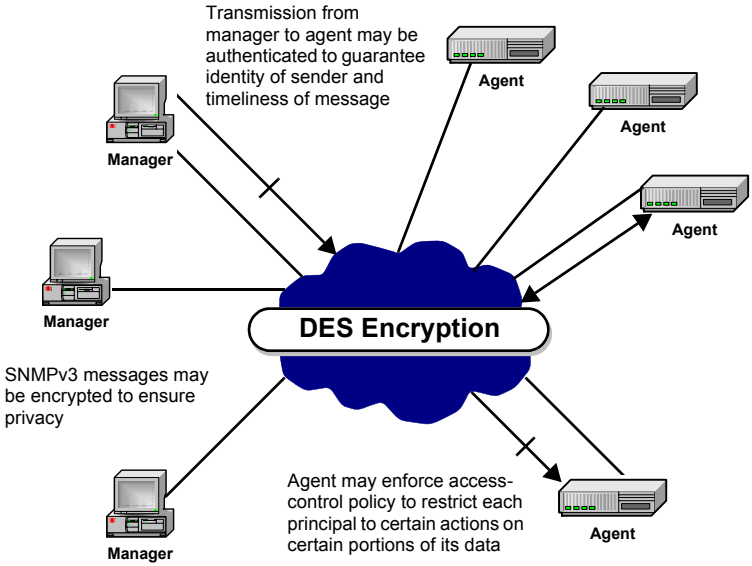
Operator	Description
get-bulk	Contrary to the get-next operation, get-bulk uses BULK Requests to query for a tree of information about a network entity. A variable put in command line specifies which portion of the object identifier space will be searched using BULK Requests. All variables in the subtree below the given variable are queried as a single request and their values presented to the user.

This message is sent from the manager to the agent.

SNMPv3

To correct the security deficiencies of SNMPv1/v2, SNMPv3 was defined with an overall SNMP architecture and a set of security capabilities. SNMPv3 includes three important services: *authentication*, *privacy*, and *access* control (Figure 4). To deliver these services in a flexible and efficient manner, SNMPv3 introduces the concept of a *principal*, which is the entity on whose behalf services are provided or processing takes place.

Figure 4: SNMPv3 Services



Definitions

Structure of Management Information (SMI)

The *Structure of Management Information* (SMI) is the set of rules for specifying the management information that a device maintains. To be more precise, the management information is actually a collection of managed objects, and these rules are used to both name and define these managed objects.

Management Information Base (MIB)

A *Management Information Base* (MIB) is a structured collection of all the managed objects maintained by a device. The managed objects are structured in the form of a hierarchical tree. At the top of the tree is the most general information available about a network. Each branch of the tree then gets more detailed into a specific network area, with the leaves of the tree as specific as the MIB can get.

Object Identifier (OID)

Object Identifiers (OID) are strings of numbers. They are allocated in a hierarchical manner, so that, for instance, the authority for "1.2.3" is the only one that can say what "1.2.3.4" means. They are used in a variety of protocols. The formal definition of OIDs comes from ITU-T recommendation X.208 (ASN.1), which is available from the ITU.

Changing a Parameter Value

Changing a parameter value involves contacting the remote unit with the Unit Manager Express. Make sure to use the MIB file that matches the version of the MIB located inside the current software build of the unit.

You must then locate the proper parameter to modify and change its value. Most of the parameters will require that you reboot the remote unit. A reboot may be software-initiated or manually initiated with the power connector or switch. It will delete all statistics stored and will overwrite all volatile parameter values in the configuration file. A reboot will also reinitiate the unit's initial provisioning sequence.

See "Performing SNMP Operations" on page 26 for more details.

CHAPTER 3

Using the Unit Manager Express

This chapter describes how to use the Unit Manager Express to read and modify MIB values for a selected remote unit.

Unit Manager Express Main Window

The Unit Manager Express main window consists of the following components:

Figure 5: Unit Manager Express Main Window

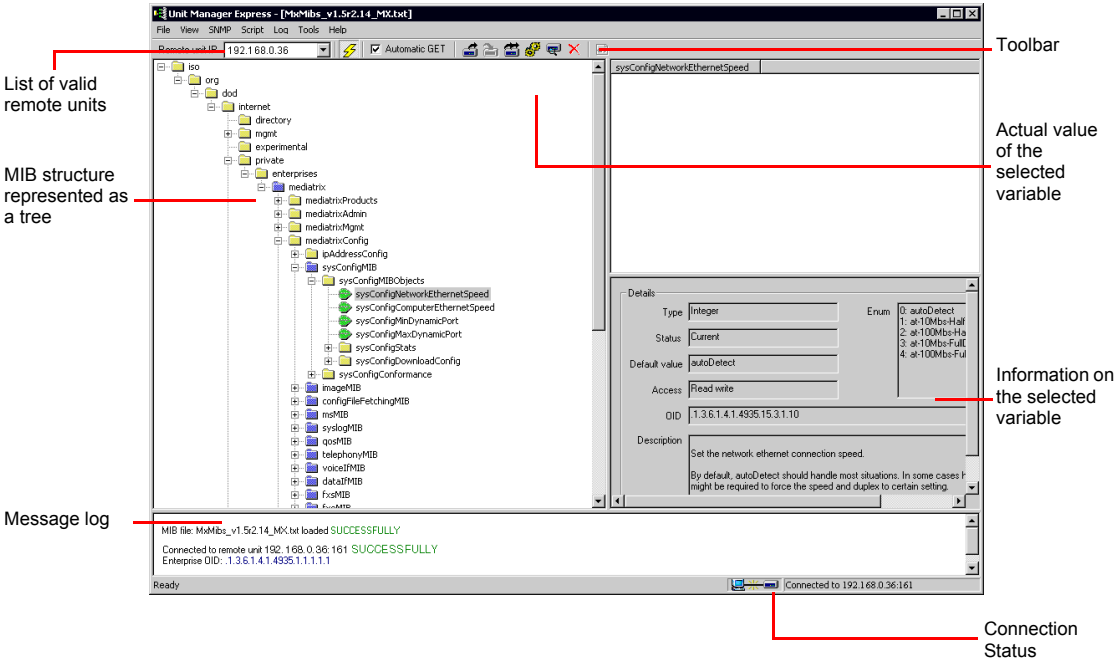


Table 4 describes each of these components.

Table 4: Unit Manager Express Components

Component	Description
List of valid remote units	Drop-down list of all valid remote unit IP addresses. See "Connecting to a Remote Unit" on page 16 for more details.
MIB structure represented as a tree	Graphical representation of the opened MIB file. See "Opening a MIB File" on page 23 for more details.

Table 4: Unit Manager Express Components (Continued)

Component	Description
Message log	Displays informative messages such as SNMP operation results, connection status, etc. See "Message Log" on page 37 for more details.
Toolbar	Tools you can use to manage the remote unit. See "Toolbar" on page 11 for more details.
Actual value of the selected variable	Clicking the button displays the actual value of the selected variable.
Variable details	Details of the selected variable. • *Type*: string, integer, etc. • *Status*: Status of the variable (*Current*, *Deprecated*, *Obsolete, etc.*). • *Default value*: value used by default. • *Access*: read, read-write, etc. • *OID*: Unique Object Identifier of the selected variable. • *Description*: Short description of the variable and its use. • *Range*: range of values accepted. May also be Index or Enum, depending on the variable type. • *Index*: In tables, displays the column name that makes up the table index. May also be Range or Enum, depending on the variable type. • *Enum*: list of choices available. May also be Range or Index, depending on the variable type.
Connection Status	Status of the connection to the selected remote unit.

Toolbar

The following tools help you manage remote units.

Table 5: Toolbar Icons

Tool	Description
	The Unit Manager Express is not connected to a remote unit. You must click this icon to connect to a selected remote unit. See "Connecting to a Remote Unit" on page 16 for more details.
	The Unit Manager Express is successfully connected to a selected remote unit. See "Connecting to a Remote Unit" on page 16 for more details.
	Performs a SNMP GET operation. See "Performing a GET Operation" on page 26 for more details.
	Performs a SNMP SET operation. See "Performing a SET Operation" on page 27 for more details.
	Performs a SNMP WALK operation. See "Performing a SNMP Walk Operation" on page 29 for more details.
	Runs an existing script. See "Running a Script" on page 41 for more details.
	Discovers remote units that run the SNMP protocol within a defined range of IP addresses. See "Autodetecting New Units" on page 17 for more details.
	Clears the content of the message log. See "Message Log" on page 37 for more details.

Setting SNMP Preferences

The Unit Manager Express can work with units that support SNMPv1, SNMPv2c, or SNMPv3.

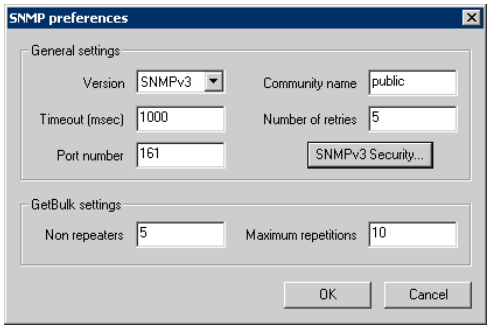
Note: SNMPv3 makes extensive use of encryption, so working with it requires a large amount of resources from your PC. Mediatrix Telecom, Inc. suggests that you use a recent PC with the appropriate CPU and memory resources.

▶ **To set SNMP preferences:**

1. Select the *SNMP:Preferences* task of the menu bar.

   The following window opens.

Figure 6: SNMP Preferences Window



2. Select the *Version* of SNMP used.

   Supported values are SNMPv1, SNMPv2c, and SNMPv3. These values are mutually exclusive. For instance, if you select SNMPv2c and you try to connect to a unit that does not support SNMPv2c, the connection will fail.

   This setting is very important because it will be used when autodiscovering units on the network. See "Autodetecting New Units" on page 17 for more details.

3. Set the *Timeout* in milliseconds.

   When a SNMP request is sent to the remote unit, an answer must be sent back from the unit within a specified period of time. This is the Timeout. If no answer is received within the Timeout value, the Unit Manager Express will send the SNMP request again to the remote unit. If the unit still does not answer after the defined Number of retries, the Unit Manager Express considers it as being off-line.

4. Define the *Port number* on which the remote unit listens for SNMP requests.

5. Define the *Community name*.

   Mediatrix Telecom, Inc. recommends not to change this value and keep *public*.

6. Define a *Number of retries*.

   Number of times the Unit Manager Express sends a SNMP request to the remote unit in case the unit does not answer within the specified Timeout. If the remote unit still does not answer after the defined Number of retries, the Unit Manager Express considers it as being off-line.

GetBulk Settings

The GetBulk operation is used to efficiently retrieve large blocks of data, such as multiple rows in a table. GetBulk settings are specific to SNMPv2c and SNMPv3. You must define these settings if you want to connect to a remote unit that supports SNMPv2c or SNMPv3.

▶ **To set GetBulk settings:**

1. Access the *SNMP preferences* window and set the *Version* to *SNMPv2c* or *SNMPv3*.

2. Set the GetBulk settings.

   The following parameters can be set:

Table 6: GetBulk Parameters

Parameter	Description
Non repeaters	Number of pairs in the variable binding list array for which a single instance should be returned.
Maximum repetitions	Maximum number of repetitions to return.

Note: *Non repeaters* and *Maximum repetitions* are SNMPv2 / SNMPv3 parameters of the get-bulk-request command. This command is not yet supported in the current version of the Unit Manager Express.

3. Click *OK* when all changes are done.

SNMPv3-Security Settings

You must set SNMPv3 security settings to successfully connect to a unit that supports SNMPv3.

Note: You must create a user in the remote unit or SNMPv3 agent prior to defining the following settings or you will not be able to connect to this remote unit or SNMPv3 agent.

Warning: The SNMPv3 method for changing the password or encryption key contains a flaw which may result in setting the incorrect password. This problem can happen if you use an incorrect "oldpassword" when changing your password. Always excercice great caution when changing your password or encryption key. Note that the factory reset can be used to clear the SNMPv3 password if your unit is using the SIP v4.4.x version.

▶ **To set SNMPv3 security settings:**

1. Access the *SNMP preferences* window and set the *Version* parameter to *SNMPv3*.
2. Click the *SNMPv3 Security* button.

   The following window opens.

Figure 7: SNMPv3 Security window

This window allows you to specify security information required to successfully connect to a SNMPv3 agent.

3. Set the following information:

Table 7: Security Settings Parameters

Parameter	Description
User name	Specifies the SNMPv3 security user name. It is a human-readable alphanumeric string representing a user or a group of users. It is passed as a parameter in all of the SNMP operations. This field cannot be empty and must have at least 8 characters. This setting is very important because it will be used when autodiscovering units on the network. See "Autodetecting New Units" on page 17 for more details.
Context name	Specifies the SNMPv3 context name. A MIB context is a named subset of the object instance in the local MIB.
Security level (radio buttons)	Specifies the SNMPv3 security level at which SNMP messages can be sent or processed, expressed in terms of whether or not authentication and / or privacy are provided. The available values are: • No authentication or privacy • Authentication without privacy • Authentication with Privacy All options are mutually exclusive.
Authentication protocol (drop-down list)	Specifies the SNMPv3 authentication protocol. Available values are *MD5* or *SHA*.
Change Password - Authentication (button)	Opens the *Change Password* window for specifying the SNMPv3 authentication password. Note that you must have the proper rights granted by the SNMPv3 agent to proceed. **Note**: SNMPv3 Passwords should not have repeating blocks of characters.
Privacy protocol (drop-down list)	Specifies the SNMPv3 privacy protocol. For example DES.
Change Password - Privacy (button)	Opens the Change Password window for specifying the SNMPv3 privacy password. Note that you must have the proper rights granted by the SNMPv3 agent to proceed. **Note**: SNMPv3 Passwords should not have repeating blocks of characters.

Table 7: Security Settings Parameters (Continued)

Parameter	Description
Save Passwords	Saves authentication and privacy passwords in the local registry.

4. Click *OK* when all changes are done.

   You will now be able to connect to a SNMPv3 agent.

Connecting to a Remote Unit

Now that you have set the SNMP preferences, you are ready to contact a remote unit that uses the SNMP protocol and start setting parameters. Please note that you must contact the remote unit before you can perform any other SNMP operation with it.

Note: Before contacting a remote unit, you must know the SNMP version it supports and set the SNMP preferences accordingly. See "Setting SNMP Preferences" on page 12 for more details.

▶ **To contact a remote unit:**

1. Type the IP address of the unit you want to contact in the *Remote Unit IP* field.

   The Unit Manager Express does not check to confirm that the IP address is valid. If the unit you want to contact is already listed in the drop-down menu, select it.

2. Click the button.

   If the operation is successful, the button replaces the button. The status bar on the bottom right will show the connected icon and a message similar to the following will be displayed in the message log:

   ```
   Connected to remote unit 192.168.0.127:161
   SUCCESSFULLY
   ```

   If the operation is not successful, the status bar on the bottom right will show the not connected icon and a message similar to the following will be displayed in the message log:

   ```
   Connection FAILED
   ```

   Once a remote unit has been contacted, the next step is to open a proper MIB file.

Autodetecting New Units

You can automatically find new units on the network that use the SNMP protocol. The process is also called discovering SNMP agents.

This process is directly affected by the SNMP version defined in the SNMP preferences of the Unit Manager Express. For instance, if SNMPv2c is selected, only devices that run SNMPv2c will be detected and listed.

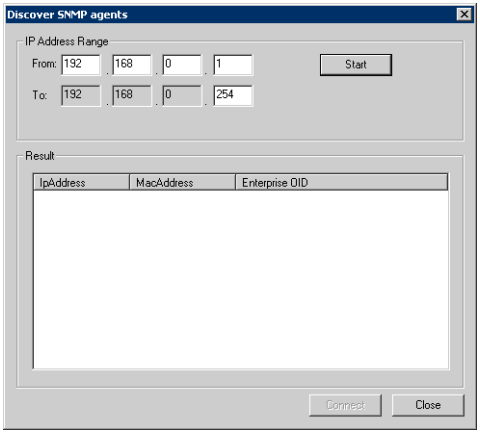
▶ **To autodetect new units on the network:**

1. Click the button.

   You can also select the *SNMP:Discover agents* option.

   The *Discover SNMP agents* window opens:

   Figure 8: Discover SNMP agents Window

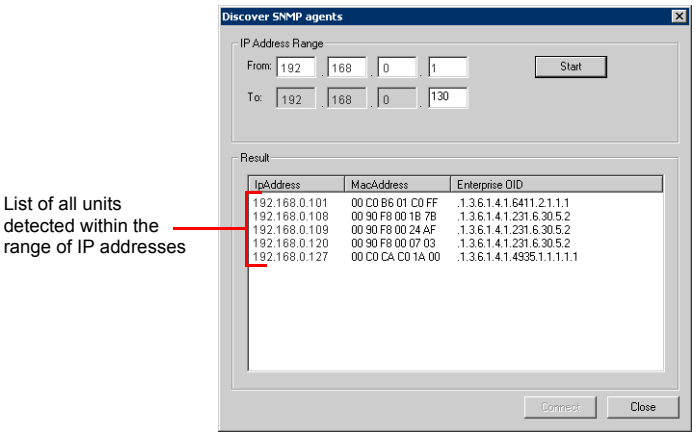


2. Set the range of IP addresses within which you want to detect units.

3. Click the *Start* button.

   The Unit Manager Express goes through all IP addresses within the specified range and lists the units detected in the *Result* section.

Figure 9: Results of Autodetect



The units are listed with their IP address, MAC address and unique enterprise OID.

4. To contact one of the units listed, double-click it.

   You can also select it and click the *Connect* button in the *Discover SNMP agents* window.

5. Click *Close* when done.

   Note that the list is not saved. The next time you open the *Discover SNMP agents* window, it will be empty.

Note: If you have defined SNMPv3 as SNMP version in the preferences, the autodetect feature will discover only SNMPv3 units that have the user name specified in the Unit Manager Express SNMP preferences.

SNMPv3 Unit Settings

You can set some SNMPv3 information on the unit or SNMP agent to which you are connected, provided this unit or SNMP agent grants you the rights to do so.

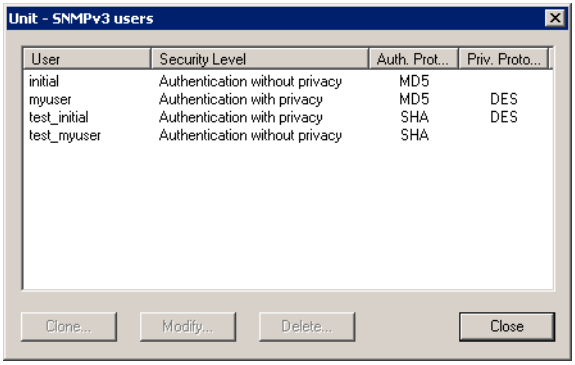
Note: The unit to which you are connected must support SNMPv3.

▶ **To set SNMPv3 information:**

1. In the *SNMP* menu, select the *SNMPv3 Unit settings* task.

   The following window opens.

Figure 10: SNMPv3 Users Window



This window lists all users currently created in the SNMP agent, including the user under which you have logged. These agents are listed with their Authentication and Privacy information.

Cloning a User

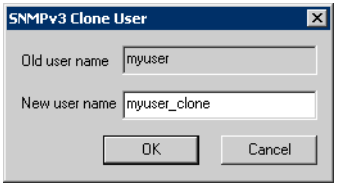
Cloning an existing user is the only way to create a new user in the SNMP agent. When cloning, all information of the original user, such as Authentication and Privacy information, is transferred to the clone.

To clone a user:

1. Select a user and click the *Clone* button.

   The following window opens:

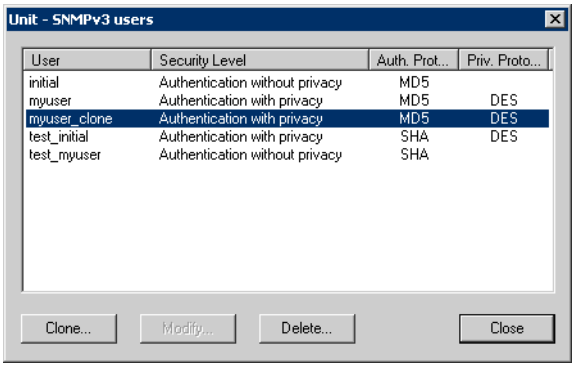
Figure 11: Clone User window



2. Type the new name of the user in the *New user name* field, then click *OK*.

   The new user is created.

Figure 12: New User Cloned



If you want to modify information on this cloned user, you will have to connect to the remote unit or SNMPv3 agent with this user. See "Setting SNMP Preferences" on page 12 for more details.

Modifying a User

You can only modify the user under which you have logged. Furthermore, the rights granted to you by the SNMPv3 agent also dictate what information you can actually modify.

▶ To modify a user:

1. Select the user under which you have logged.

   The *Modify* button becomes available.

2. Click the *Modify* button.

   The following window opens.

Figure 13: SNMPv3 Security Window

   Some settings cannot be modified.

3. If applicable, modify the *Group name*.

   The Group name specifies the SNMPv3 group name in which the user is located.

4. Select a Security level.

   This is the SNMPv3 security level at which SNMP messages can be sent or processed, expressed in terms of whether or not authentication and / or privacy are provided. The available values are:

   - No authentication or privacy
   - Authentication without privacy
   - Authentication with privacy

All options are mutually exclusive. To change the security level, you must choose the group corresponding to the security level selected (in the *Group name* field):

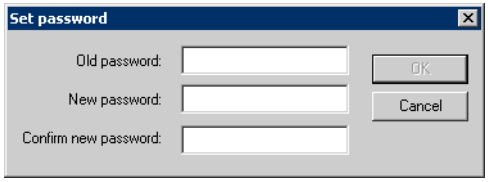
- For *Authentication with privacy*, select the *AuthPrivGrp* group.
- For Authentication without privacy, select the *AuthNoPrivGrp* group.
- For *No authentication or privacy*, select the *NoAuthNoPrivGrp* group.

Note that you can only select options that are less secure, for instance, you cannot go from *Authentication without privacy* to *Authentication with privacy*. Also, you cannot change the Authentication or Privacy protocol.

5. If applicable, click the *Change Password* button of either the *Authentication* or *Privacy* section.

   The following window opens:

Figure 14: Set password window



You can modify the password by supplying the old password and entering a new one.

Caution: SNMPv3 Passwords should not have repeating blocks of characters.

Deleting a User

You can delete an existing user.

▶ **To delete a user:**

1. Select the user you want to delete.
2. Click the *Delete* button.

   The selected user is removed from the list.

Opening a MIB File

The MIB file contains the actual variables you will be able to set for the selected remote unit. By default, the Unit Manager Express is not installed with any MIB files. You can install the MIB file you require as described in "Installing a MIB File" on page 3. The MIB file will be automatically opened upon starting the Unit Manager Express. You can also copy the MIB file you want in a directory. See "Notes on MIB files" on page 24 for more details.

When opening a MIB, you must ensure that it matches the MIB of the remote unit, or you will not be able to perform GET and SET operations.

A good way to know if you have the proper MIB file is to perform a GET operation on a variable. If you get something like the following, this probably means you have not opened the proper MIB file.

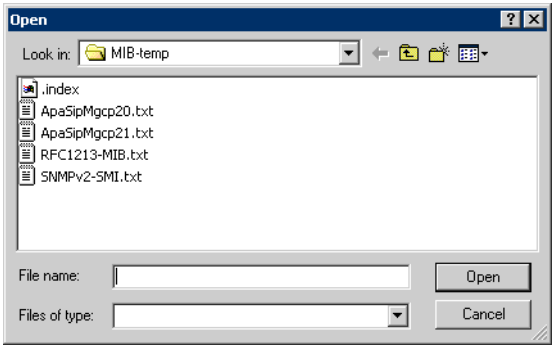
```
Get request result: SNMP ERROR
There is no such variable name in this MIB
```

▶ **To open a MIB file:**

1. In the *File* menu, select the *Open* task.

   The following window opens.

Figure 15: Open Window



2. Select the proper MIB file. Ensure that it matches the MIB file in the remote unit you have contacted.

   The MIB structure is displayed in the left part of the main Unit Manager Express window. You can browse through this structure.

MIB File Icons The MIB information is represented by icons as described in Table 8.

Table 8: MIB Structure Icons

Icon	Description
	Group
	Module
	Variable – also called a leaf
	Table
	Table Entry
	Column in a table
	Index column in a table
	Trap

Notes on MIB files

- There are two types of MIB files: standard and unit-specific.

  The standard MIB files are *RFC1213-MIB.txt* and *SNMPv2-SMI.txt*. These MIBs contain a standard structure that must be present in your MIB structure.

  The unit-specific MIB files can be anything you want. However, these MIB files must contain all the modules that make up your MIB structure. Let's say for instance your MIB structure contains three (3) MIB modules: Module A, Module B and Module C. Each of these modules addresses specific parameter settings. Your MIB text file must contain each of these modules, or you will not be able to open the MIB structure.

- All your MIB files must be located in the same directory.

  The standard MIBs must be located in the same directory as the unit-specific MIBs. If this is not the case, the following error message will be displayed and the MIB file will not open.

  ```
  cannot find module SNMPv2-SMI
  ```

Mediatrix Telecom, Inc. suggests that you save all your MIB files in the *Mibs* directory under the *Unit Manager Express* installation directory.

- Mx Experimental MIBs

  The Unit Manager Express does not support MIBs that are located under the *mediatrixExperimental* branch of the MIB structure because it does not have specific tasks to manage variables in experimental MIBs.

  The *mediatrixExperimental* branch is the area where objects and events in MIBs under development can be placed without fear of conflicting with other MIBs. When the items rooted under an experimental sub-tree are ready for release, they will be under a permanent branch.

  Even though experimental MIBs can be viewed with Mediatrix Telecom, Inc. configuration tools, SNMP operations may not work properly on them.

  Refer to the *MIB Reference Manual* for more details.

Performing SNMP Operations

Once a remote unit has been successfully contacted and a proper MIB file opened, you can perform SNMP operations on this unit. The Unit Manager Express allows you to perform the following SNMP operations:

- GET
- SET
- WALK

Performing a GET Operation

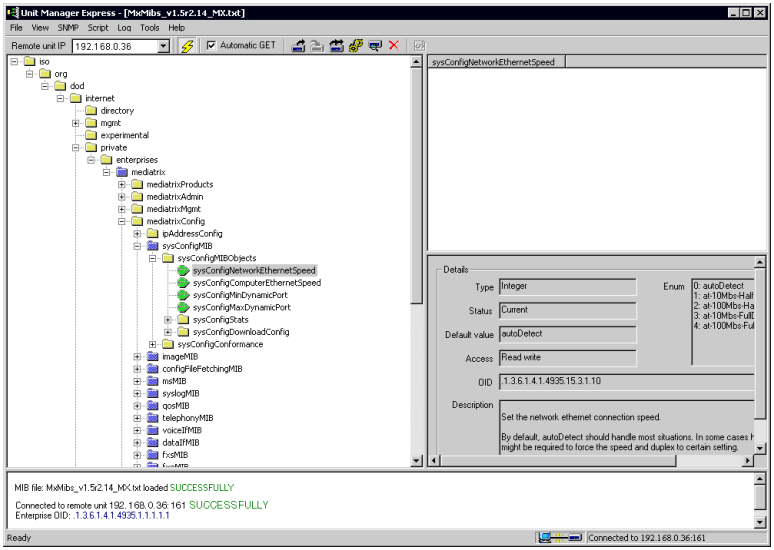
The GET operation allows you to poll the actual value of the selected variable.

Performing a GET operation:

1. Select the variable you want to poll by expanding and collapsing tree branches as required with the [+] and [-] icons.

   The variable details are displayed in the corresponding fields on the right. Figure 16 on page 26 illustrates a selected variable.

Figure 16: Selecting a Variable



2. Click the button to display the current value of the selected variable.

   You can also:

   - right-click the variable and select the *Get* option in the context-sensitive menu that appears
   - select the *SNMP:Get* option in the menu bar

   The value is displayed in the Actual value section as well as in the message log section.

Automatic GET

You can specify that the Unit Manager Express automatically GETs the value of a variable when browsing through the MIB by checking the *Automatic GET* option in the main window. The Unit Manager Express will automatically send a GET request result in the message log and the variable's value will be displayed in the Actual value section.

Performing a SET Operation

The SET operation allows you to modify the value of the selected variable.

▶ **To modify (SET) a variable value:**

1. Select the variable you want to modify by expanding and collapsing tree branches as required with the [+] and [-] icons.

   The variable details are displayed in the corresponding fields on the right.

2. Click the button to display the current value of the selected variable.

   You can also:

   - right-click the variable and select the *Get* option in the context-sensitive menu that appears
   - select the *SNMP:Get* option in the menu bar

   The value is displayed in the Actual value section. If you have enabled the *Automatic GET* option, the value is displayed when selecting the variable.

3. Select the value in the Actual value section.

   This value becomes highlighted.

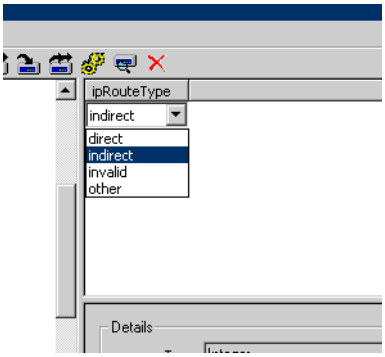
4. Modify the value for the variable, then click the button.

   You can also:

   - right-click the variable and select the *Set* option in the context-sensitive menu that appears
   - select the *SNMP Query:Set* option in the menu bar
   - press the <Enter> key of your keyboard

   If the value type is *Enum*, you can click twice on the variable at a one second interval and a drop-down menu with all the available values will be displayed. Select the proper value.

Figure 17: Drop-Down Menu



   Make sure that:

   - The variable is not read-only
   - The value you are setting falls within the range of accepted values

   You can apply the same value to all interfaces at once (for instance, to all the ports) by right-clicking the value and selecting *Apply value to all* in the context-sensitive menu that appears.

Performing a SNMP Walk Operation

The SNMP Walk operation is usually performed on a node (group). It repeatedly queries the remote unit by using the GETNEXT operation.

▶ To perform a SNMP Walk operation:

1. Select a node (group) for which you want to get the value of all sub-groups and variables.

   Expand and collapse tree branches as required to find the group you want to modify.

Note: Performing a SNMP Walk operation on one variable is the same as performing a SNMP GET operation on this variable.

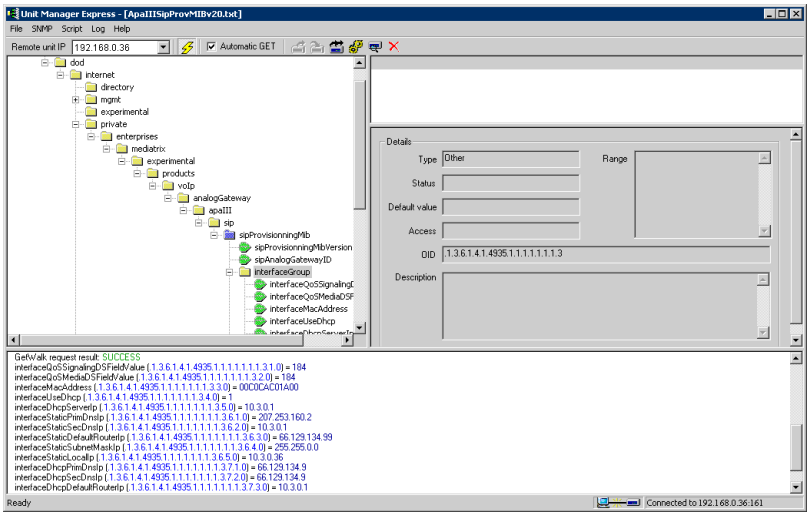
2. Click the button.

   You can also:

   - right-click the node and select the *GetWalk* option in the context-sensitive menu that appears
   - select the *SNMP:GetWalk* option in the menu bar

   The value of all sub-groups and variables is listed in the message log.

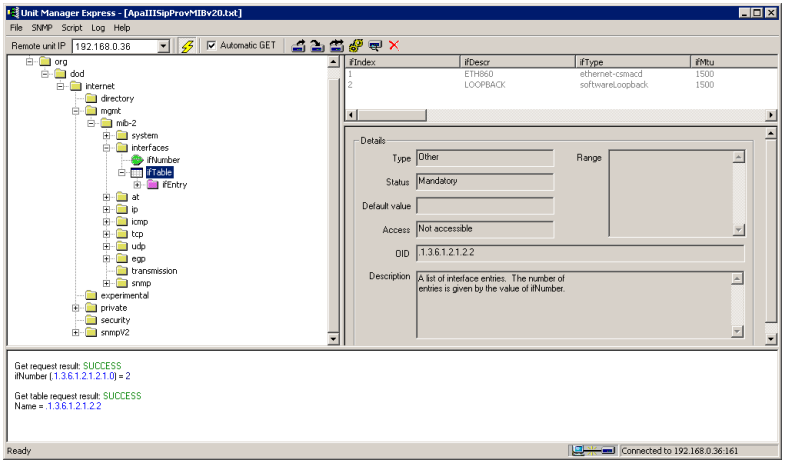
Figure 18: SNMP Walk Operation



SNMP Table Viewer

The Unit Manager Express has an integrated table viewer to display the tables in a MIB file. The following figure illustrates a table in a MIB file.

Figure 19: SNMP Table Viewer



Options

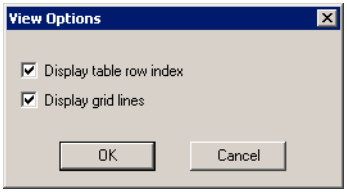
The Unit Manager Express offers you the possibility to show or hide the table row index and the table grid lines.

▶ **To change table viewing options:**

1. In the *View* menu, select the *Options* task.

   The *View Options* window opens.

Figure 20: View Options Window



2. Select if you want to show/hide the table row index and grid lines by checking the proper choices.

3. Click *OK* to set the changes.

The changes will be displayed on screen the next time you select a table and GET its values.

Figure 21: Table Viewing Options Examples

Options off

ifIndex	ifDescr	ifType
1	ETH860	ethernet-csmacd
2	LOOPBACK	softwareLoopback

Options on

	ifIndex	ifDescr	ifType
1	1	ETH860	ethernet-csmacd
2	2	LOOPBACK	softwareLoopback

Table row index Table grid lines

Performing a GET Operation

You can perform a SNMP GET operation as described in "Performing a GET Operation" on page 26. However, you can perform two types of GET operations:

- On the Table itself: all values of the table are displayed in the Actual value section.
- On a column of the Table: all values in this column are displayed in the Actual value section.

Performing a SET Operation

You can perform a SNMP SET operation as described in "Performing a SET Operation" on page 27. This SET operation is performed on a value (cell) of the table.

Note: Before making any change in a table, make sure the new value falls within the range of the variable and that the change is required.

You can apply the value of a row to all other rows in the table by right-clicking the value and selecting *Apply row to all* in the context-sensitive menu that appears.

Performing a Walk Operation

You can perform a SNMP WALK operation as described in "Performing a SNMP Walk Operation" on page 29. However, you can perform two types of WALK operations:

- On the Table itself: all values of the table are displayed in the Actual value section.
- On a column of the Table: all values in this column are displayed in the Actual value section.

Generating a XML Configuration File

You can generate a XML file that will contain the configuration of the unit connected to the Unit Manager Express.

Preferences

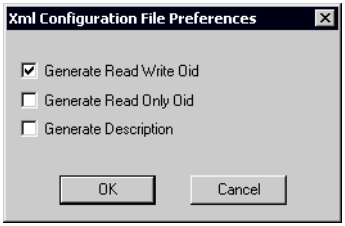
The XML configuration file preferences define the information that must be in the generated configuration file.

▶ **To set the XML configuration file preferences:**

1. In the *Tools* menu, select the *Xml Configuration File Preferences* task.

   The *Xml Configuration File Preferences* window opens.

Figure 22: Xml Configuration File Preferences Window



2. Select which parameter(s) you want to include in the XML configuration files.

Table 9: XML Preferences

Parameter	Description
Generate Read Write OID	The Read/Write OIDs in the MIB tree are included in the configuration file.
Generate Read Only OID	The Read-Only OIDs in the MIB tree are included in the configuration file.
Generate Description	The description of all the OIDs in the MIB tree are included in the configuration file. **Note**: This option may create a much larger file.

3. Click *OK*.

Generating the File

Once the preferences are set, you can generate the configuration file.

▶ **To generate a XML configuration file:**

1. In the *Tools* menu, select the *Generate Xml Configuration File* task.
2. Select the name and location of the configuration file.

   The Unit Manager Express generates each of the parameters in the MIB tree according to the preferences you have set and the contents of the initialization file (see "Initialization File" on page 34 for more details).

   As an alternative, you can also right-click anywhere in the MIB tree and select the *Generate Xml Configuration File* option in the context-sensitive menu that opens.

   In this case, only the OID under the selected element will be included in the configuration file according to the preferences you have set and the contents of the initialization file.

Initialization File

The initialization file defines how to generate the configuration file. This file, called *ConfigFile.ini*, is located under the */Mibs* sub-directory.

Mediatrix Telecom, Inc. recommends that you proceed with care if you decide to modify this file because it could prevent you from properly generating an XML configuration file.

The following is the format of the initialization file:

```
[Exclude OID]
OID=1.3.6.1.2.1.4.1.0

[Include RO OID]
OID=1.3.6.1.2.1.4.12.0

[Include Not Mediatrix Subtree]
OID=1.3.6.1.2.1.10.23

[Modules Order]
Module=MX-SIP-DEBUG-MIB
Module=MX-DEBUG-MIB

[Comment OID]
1.3.6.1.4.1.4935.15.3.1.30.1.0=(The following ...)
1.3.6.1.4.1.4935.99.23.1.5.0=(The following ...)

[Comment Group OID]
1.3.6.1.4.1.4935.15.40.1.200=(The following 3 ...)
1.3.6.1.4.1.4935.99.10.1.5=(The following 2 ...)
1.3.6.1.4.1.4935.15.120.1.250=(The following ...)

[Comment Module]
MX-DEBUG-MIB=(The following 7 variables are ...)
```

The following table describes the various sections of this file.

Table 10: Initialization File

Section	Description
Exclude OID	The OIDs listed under this section are excluded from the generated file.
Include RO OID	The Read-Only OIDs listed under this section are included in the configuration file, even if the *Generate Read Only OID* preference is not selected.
Include Not Mediatrix Subtree	The OIDs and sub-OIDs, if any, that are not under the Mediatrix tree are included in the generated file.

Table 10: Initialization File (Continued)

Section	Description
Modules Order	The modules listed under this section are the first to be generated in the order given in the generated file.
Comment Oid	The comments listed under this section are included in the generated file above their corresponding OID. For a variable in a table, the OID is the one of the column.
Comment Group Oid	The comments listed under this section are included in the generated file above their corresponding group OID. **Note**: For the values in a table, the Comment Group OID acts as a comment on the table, you must thus list the OID of the table to generate a comment above it.
Comment Module	The listed under this section are included in the generated file above their corresponding module.

Miscellaneous Options

The following options are very useful when working with a MIB.

Using the Find Option

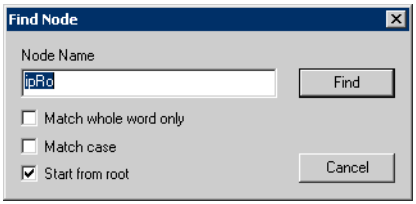
You can use the *Find* option to look for a variable name or a string of characters.

▶ **To use the find option:**

1. Right-click in the MIB tree where you want the search to start.
2. Select *Find* in the context sensitive menu that appears.

   The following window opens:

Figure 23: Find Node Window



3. Type the variable name or character string you want to find in the *Node Name* field.
4. Select one or more of the following options:
   - Match whole word only
   - Match case
   - Start from root

   It is recommended to always check the *Start from root* option, because the Find option only works in a descending way.
5. Click *Find* when ready.

Expanding and Collapsing the MIB Tree

You can expand or collapse all of the MIB tree by right-clicking anywhere in the MIB structure and selecting either *Expand* or *Collapse* in the context sensitive menu that appears.

Message Log

The message log section displays informative messages that are generated when working with a remote unit or the opened MIB. These messages may include connection status, SNMP operation results, etc.

The SNMP operation results message will have the following syntax (if successful):

```
Get/Set/GetWalk request result:SUCCESS
VariableName(OID)=value
```

For example:

```
Get request result: SUCCESS
sysUpTime (.1.3.6.1.2.1.1.3.0) = 7:17:35:27.00
```

You can clear the message log window by clicking the button or selecting the *Log:Clear* option of the menu bar.

CHAPTER 4

Using Scripts

This chapter describes how to use and create scripts to manage remote units.

Recording a Script

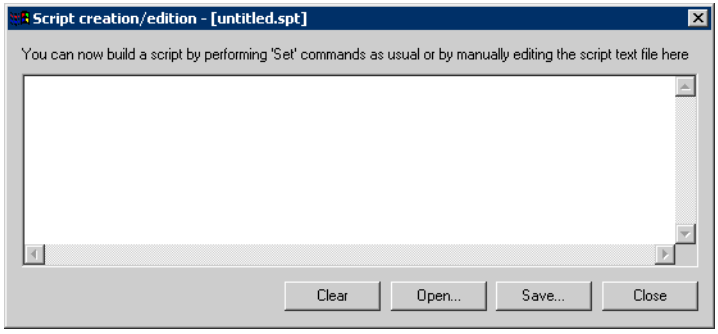
You can create a script that will contain several SNMP SET operations. You will then be able to use this script on any number of remote units that have the same MIB, thus saving time.

▶ **To record a script:**

1. In the *Script* menu, select the *Create/Edit* option.

   The *Script creation/edition* window opens.

Figure 24: Script creation/edition Window

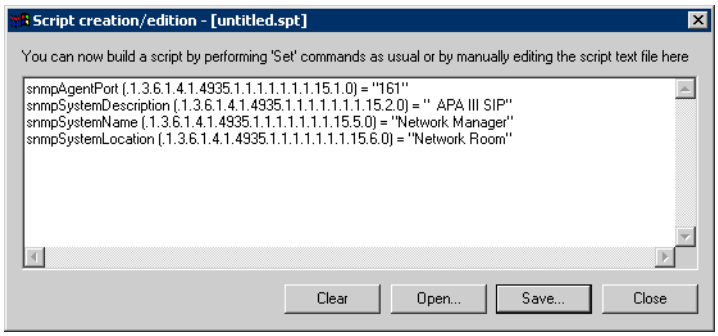


2. While leaving the window opened, perform the SNMP SET operations you want to record.

   Each successful SET operation is displayed in the *Script creation/edition* window. If the operation is not successful, it is discarded.

   There is no limit to the number of operations that you can record.

Figure 25: Script Operations



3. When you have all the operations you want, save the script by clicking the *Save* button of the *Script creation/edition* window.

   Mediatrix Telecom, Inc. suggests that you save all your scripts in the *Scripts* directory under the *Unit Manager Express* installation directory.

   The script is now ready to be used.

Running a Script

You can run an existing script on a remote unit. The Unit Manager Express must be properly connected to this remote unit. See "Connecting to a Remote Unit" on page 16 for more details.

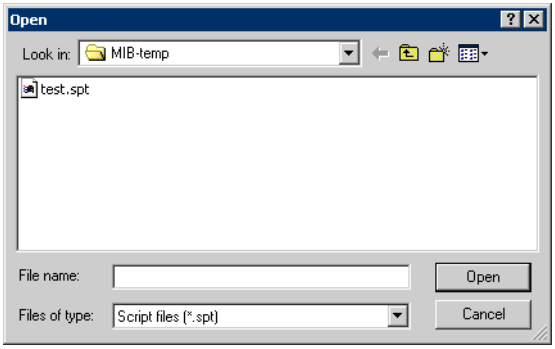
▶ **To run a script:**

1. Click the button.

   You can also select the *Script:Run* option.

   The following window opens.

Figure 26: Script Open Window



2. Select the script you want to run and click *Open*.

   The script starts and the SET operations it contains are performed. The operations, possible error messages, and result are listed in the message log window.

Note: If the script contains syntax errors, the execution is cancelled.

3. If you want to run the script on another unit, connect to this unit and repeat steps 1-2.

Editing a Script

You can edit an existing script by setting new SET operations or manually editing the script text file.

▶ **To edit an existing script:**

1. In the *Script* menu, select the *Create/Edit* option.

   The *Script creation/edition* window opens.

2. Click the *Open* button.

   The script selection window opens.

3. Select the script you want to edit and click the *Open* button.

   The script is displayed in the *Script creation/edition* window.

4. Perform the modifications you want to make.

   You can:

   - Clear the content of the screen by clicking the *Clear* button.
   - Add new SET operations by performing them. See "Recording a Script" on page 39 for more details.
   - Manually edit the script by cutting, pasting, deleting or typing text. The *Script creation/edition* window is a small text editor that supports all basic functions.

   If you want to manually edit the script, each entry must follow this syntax:

   ```
   VariableName (OID) = "value"
   ```

   For example, a valid command could be:

   ```
   snmpSystemContact (.1.3.6.1.4.1.4935.1.1.1.1.1.1.
   1.15.4.0) = "Network Manager"
   ```

   A "*" is displayed beside the file name to show that changes have been done to the file.

Note: You can remove the '= *"value"*' part. The operation then becomes a GET of the specified variable instead of a SET.

5. When the modification is done, click the *Save* button.

   The script is saved and ready to be used.

Generating a Script

You can generate a script from the result of a GETWALK operation of the connected unit, for instance.

▶ **To generate a script:**

1. In the *Script* menu, select the *Generate UME Script* option.
2. Select the name and location of the script.

   The Unit Manager Express generates a script with all the Read/Write parameters in the Mediatrix MIB tree.

APPENDIX

A Glossary

Internet Protocol (IP) The IP protocol is a standard describing software that keeps track of the Internet's addresses for different nodes, routes outgoing messages, and recognises incoming messages.

Local Area Network (LAN) Data-only communications network confined to a limited geographic area, with moderate to high data rates. See also WAN.

Media Access Control (MAC) Address A layer 2 address, 6 bytes long, associated with a particular network device; used to identify devices in a network; also called hardware or physical address.

Network A group of computers, terminals, and other devices and the hardware and software that enable them to exchange data and share resources over short or long distances. A network can consist of any combination of local area networks (LAN) or wide area networks (WAN).

Object Identifier (OID) Object identifiers are strings of numbers. They are allocated in a hierarchical manner, so that, for instance, the authority for "1.2.3" is the only one that can say what "1.2.3.4" means. They are used in a variety of protocols. The formal definition of OIDs comes from ITU-T recommendation X.208 (ASN.1), which is available from the ITU.

Protocol A formal set of rules developed by international standards bodies, LAN equipment vendors, or groups governing the format, control, and timing of network communications. A set of conventions dealing with transmissions between two systems. Typically defines how to implement a group of services in one or two layers of the OSI reference model. Protocols can describe low-level details of machine-to-machine interfaces or high-level exchanges between allocation programs

Simple Network Monitor Protocol (SNMP) SNMP is the protocol governing network management and the monitoring of network devices and their functions.

Subnet Means of splitting packets into two fields to separate packets for local destinations from packets for remote destinations in TCP/IP networks. This makes small networks more efficient.

Telephony The science of translating sound into electrical signals, transmitting them, and then converting them back into sound.

Transmission Control Protocol/Internet Protocol (TCP/IP) A suite of communications protocols developed by the Department of Defense in the 1970s that connects hosts on the Internet and provides the standards for transmitting data over networks.

User Datagram Protocol (UDP) An efficient but unreliable, connectionless protocol that is layered over IP, as is TCP. Application programs are needed to supplement the protocol to provide error processing and retransmission of data. UDP is an OSI layer 4 protocol.

Voice Over IP (VoIP) The technology used to transmit voice conversations over a data network using the Internet Protocol. Such data network may be the Internet or a corporate Intranet.

Wide Area Network (WAN) A large (geographically dispersed) network, usually constructed with serial lines, that covers a large geographic area. A WAN connects LANs using transmission lines provided by a common carrier.

APPENDIX B

List of Acronyms

DES	Data Encryption Standard
IP	Internet Protocol
LAN	Local Area Network
MAC	Media Access Control
MIB	Management Information Base
OID	Object Identifier
PBX	Private Branch eXchange
PSTN	Public Switched Telephone Network
SCN	Switched Circuit Network
SHA	Secure Hash Algorithm
SMI	Structure of Management Information
SNMP	Simple Network Management Protocol
TCP/IP	Transmission Control Protocol/Internet Protocol
UDP	User Datagram Protocol
VoIP	Voice over Internet Protocol
WAN	Wide Area Network
XML	eXtensible Markup Language

Index

A

acronyms 47
authentication, in SNMPv3 15
autodetecting a remote unit 17

C

collapsing, the MIB tree 36
configuration file, generating in XML 32, 33
connecting to a remote unit 16

D

discovering SNMP agents 17

E

end user technical support viii
expanding, the MIB tree 36

F

find, option 36

G

GET operation 26, 31
    automatic 27
GetBulk, SNMP preferences 13

I

installing
    MIB file 3
    Unit Manager Express 1
intended audience v

M

main window
    actual variable value 10
    connection status 10
    list of valid remote units 9
    message log 10
    MIB structure 9
    toolbar 10
    variable details 10
message log 37
MIB
    defined 7
    icons 24
MIB (*continued*)
    in SNMP protocol 5
    installing 3
    OID 8
        in autodetect 17
        in message logs 37
        variable details 10
    opening a file 23
    SMI 7
    SNMP messages 6

O

OID
    defined 8
    in autodetect 17
    in message logs 37
    variable details 10
opening a MIB file 23

P

privacy, in SNMPv3 15

R

remote unit
    autodetecting 17
    connecting to 16

S

script
    editing 42
    generating 43
    recording 39
    running 41
security level, in SNMPv3 15
SET operation 27, 31
SMI, defined 7
SNMP
    defined 5
    MIB 5
    operations
        GET 26, 31
            automatic 27
        SET 27, 31
        Walk 29, 31
    preferences
        community name 13
        GetBulk settings 13

SNMP (*continued*)
preferences
maximum repetitions 13
non repeaters 13
number of retries 13
port number 13
SNMPv3
authentication 15
context name 15
privacy 15
security level 15
user name 15
timeout 12
version 12
SNMPv3 settings
cloning a user 20
deleting a user 22
modifying a user 21

T

table viewer 30
options 30
technical support for end user viii
toolbar icons
Clear Log 11
Connected 11
Discover Agents 11
GET 11
Not connected 11
Run Script 11
SET 11
WALK 11

U

uninstalling 4
using this manual vi

W

Walk operation 29, 31
what's new in this version v

X

XML configuration file
generating 32, 33
preferences 32